MAE1001

Fall 2026

Screwdriver Disassembly

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Course webpage: https://gwu-mae1001.github.io/fall2026/

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Learning Objectives

1

Reverse Engineering

Take apart a cordless screwdriver and identify how it works.

2

Core Engineering Concepts

Understand gears and motors, and how they create rotational power (torque).

3

Ethical AI

Apply AI responsibly to understand concepts and equations — and verify what it tells you.

Concepts I: Core Engineering

Torque

Rotational force

$$\tau = \text{Force} \times \text{distance}$$

Power

Rate at which work is done or energy is transferred

$$P = \text{Voltage} \times \text{Current}$$

Efficiency

How much input power becomes useful output

$$\eta = (P_{out} \div P_{in}) \times 100\%$$

Gear Teeth

The number of teeth on a gear

$$n = \text{tooth count}$$

Gear Ratio

Teeth relationship between two meshing gears; sets the speed–torque tradeoff. Higher ratio = more torque, less speed

$$GR = n_{driven} \div n_{driver}$$

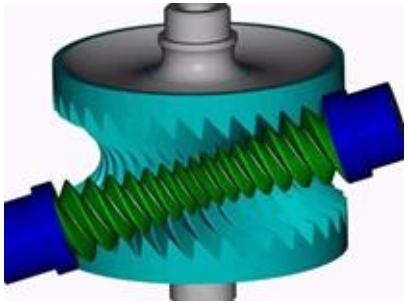
Concepts II: Types of Gears

Spur Gear



Simple, cost-effective gears with straight teeth that transmit motion and power between two parallel shafts.

Worm Gear



A screw-shaped gear (the worm) rotates and meshes with a toothed wheel (the worm wheel) at 90°.

Planetary Gear



A sun gear in the middle, multiple planet gears revolving around it, and a ring gear at the edge.

Key Components

1

DC Motor

Converts electrical energy into mechanical rotational energy

2

Planetary Gears

Multiply torque while reducing speed (and vice versa)

3

Trigger Mechanism

Controls the power input

4

Battery & Circuit Board

Supply and regulate electricity

Discuss

Why are planetary gears preferred in screwdrivers?

Lab Instructions

Materials Needed

- One screwdriver per 2 students
- Phone to take pictures of each step
- Laptop to create a lab notebook recording the lab

For each step: add a picture and add your observations.

Heads up

Questions will be asked for the Exit Ticket — keep careful notes.

Lab Steps

1. Remove the battery pack
2. Remove the security clip and take the two sections apart
3. Open the orange casing to reveal the motor
4. Remove all components from inside the head to reveal the gears

Goal

Now put it back together









Questions: Add to Lab Notebook



Motor Gear



Gear

1. How many gear teeth are on the Motor Gear and the Gear?

2. What is the gear ratio between the Motor Gear and the Gear?

3. Use AI to answer the following (show your prompt and the answer):

   a. Solve the ring-gear-fixed gear ratio given these tooth counts: Sun 10, Planet 20, Ring 50.

   Ask the chatbot to identify all the parts

Deliverables

10/30	**Lab**	Complete the lab in class.	**We're here!**
12/05	**Lab Notebook**	Submit a lab notebook showing each step taken, an image of each step, and your observations and description.	
12/05	**Ethical AI**	Answer the questions about gears using AI (include your prompts).	

Any questions? Ask away!

Come to Office Hours

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